

ROSEBANK INTERNATIONAL

Student Number: ST10454360

Portfolio of Evidence: Part 1

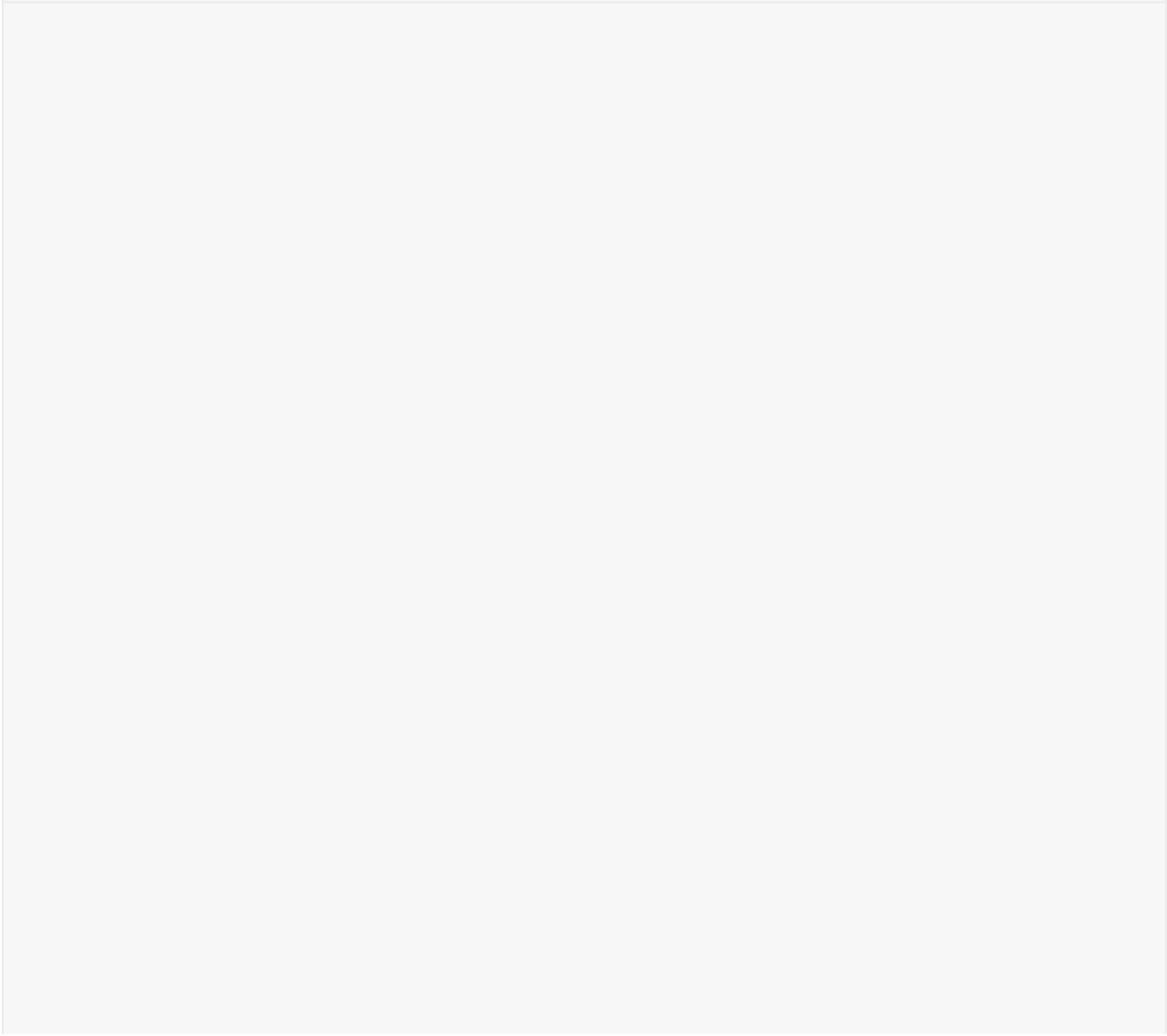
Due Date: 04 September 2026

Module: PROG6212

SECTION A

The following ERD represents the RaceDay database design. It includes the primary entities, attributes, primary keys, foreign keys, and relationships required for the system.

mermaid



USERS		
int	UserId	PK
nvarchar	FirstName	
nvarchar	LastName	
nvarchar	Email	
nvarchar	PasswordHash	
nvarchar	Role	
nvarchar	PhoneNumber	
datetime2	CreatedAt	

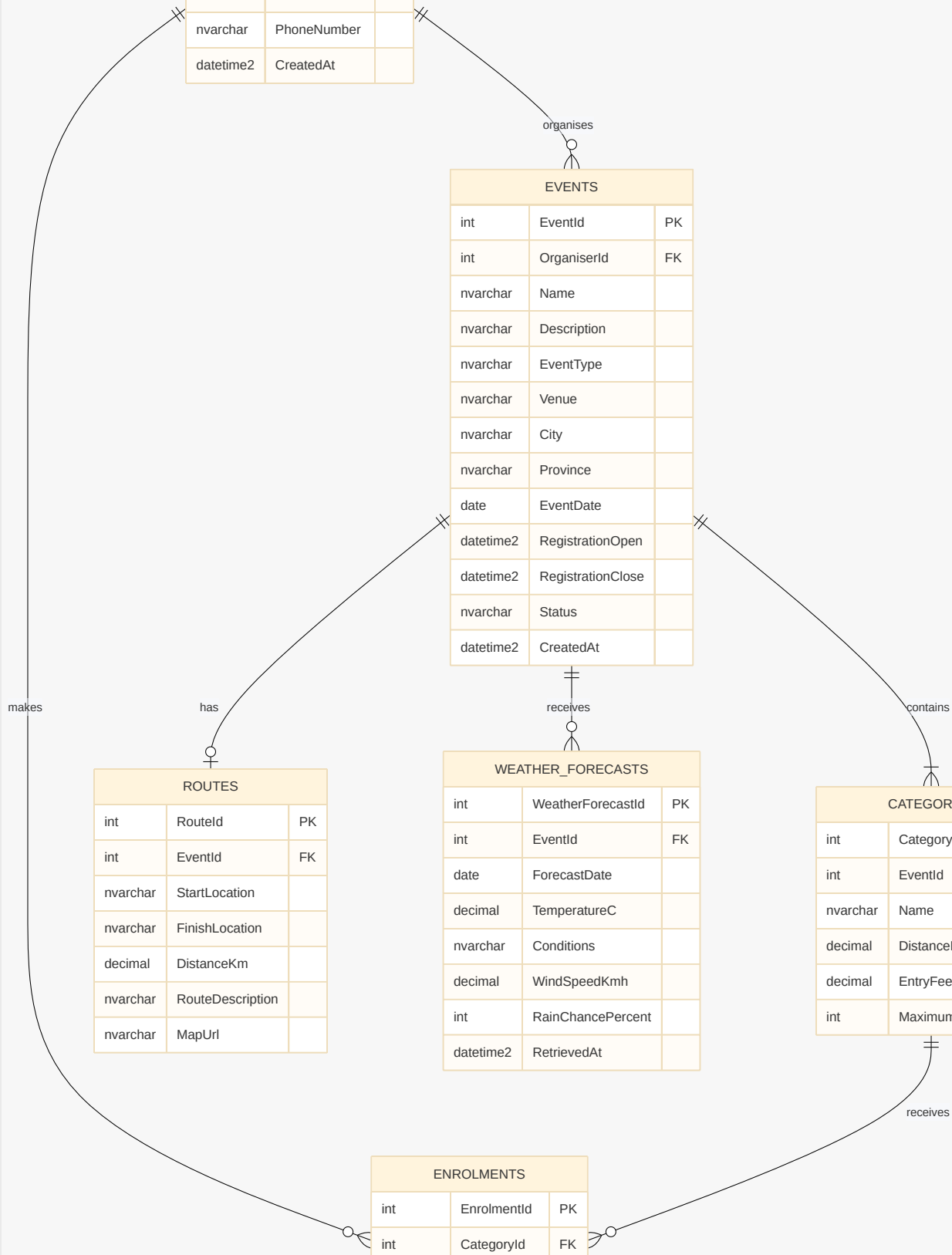
EVENTS		
int	EventId	PK
int	OrganiserId	FK
nvarchar	Name	
nvarchar	Description	
nvarchar	EventType	
nvarchar	Venue	
nvarchar	City	
nvarchar	Province	
date	EventDate	
datetime2	RegistrationOpen	
datetime2	RegistrationClose	
nvarchar	Status	
datetime2	CreatedAt	

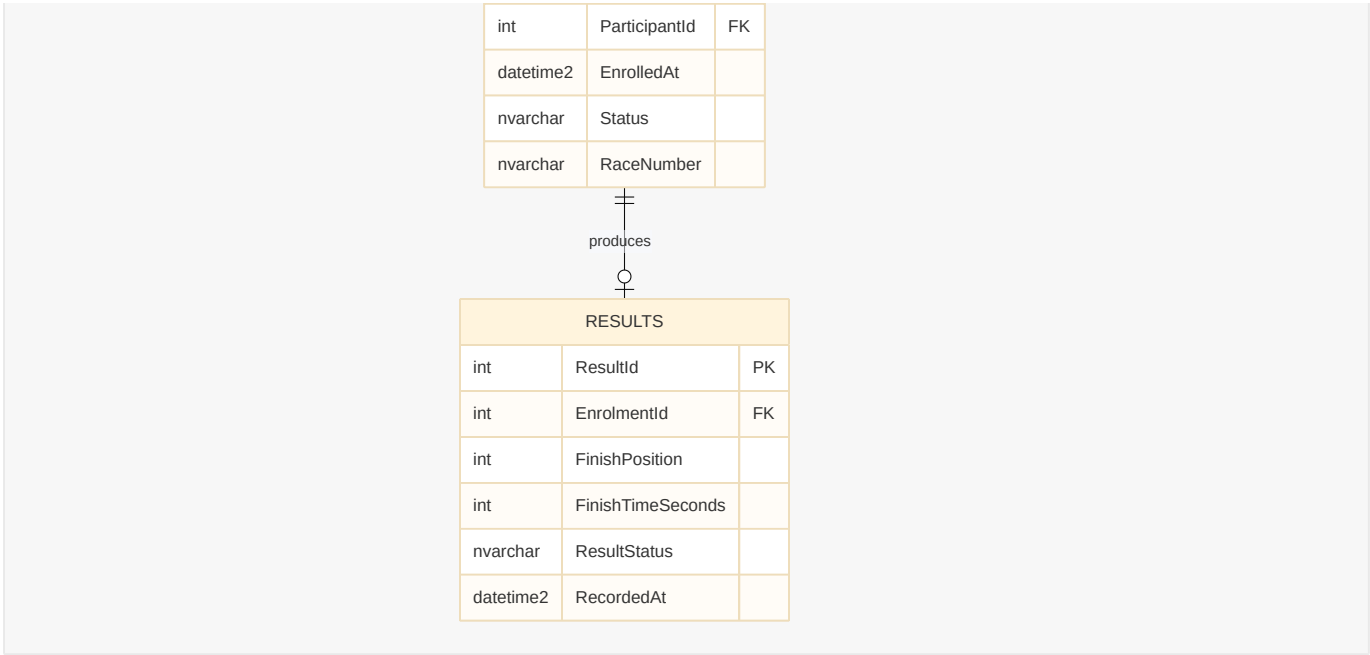
ROUTES		
int	RouteId	PK
int	EventId	FK
nvarchar	StartLocation	
nvarchar	FinishLocation	
decimal	DistanceKm	
nvarchar	RouteDescription	
nvarchar	MapUrl	

WEATHER_FORECASTS		
int	WeatherForecastId	PK
int	EventId	FK
date	ForecastDate	
decimal	TemperatureC	
nvarchar	Conditions	
decimal	WindSpeedKmh	
int	RainChancePercent	
datetime2	RetrievedAt	

CATEGORIES		
int	CategoryId	PK
int	EventId	FK
nvarchar	Name	
decimal	DistanceKm	
decimal	EntryFee	
int	MaximumEntries	

ENROLMENTS		
int	EnrolmentId	PK
int	CategoryId	FK





ERD Relationship Summary

Relationship	Cardinality	Explanation
Users to Events	One-to-many	One organiser can organise many events.
Events to Categories	One-to-many	One event can contain many categories.
Users to Enrolments	One-to-many	One participant can have many enrolments.
Categories to Enrolments	One-to-many	One category can have many participant enrolments.
Enrolments to Results	One-to-zero-or-one	An enrolment may have no result until the race is completed.
Events to Routes	One-to-zero-or-one	An event may have one route.
Events to Weather Forecasts	One-to-many	An event may have several forecasts for different dates.

SECTION B — API ENDPOINT PLAN

B.1 Purpose of the API

The RaceDay API will allow participants, event organisers, administrators, and public users to interact with the system. The API will provide access to authentication, user profiles, events, categories, enrolments, results, route information, and weather information.

All responses will be returned in JSON format. Authentication-protected endpoints will use a bearer token. Passwords must never be returned in API responses.

B.2 User Roles

Role	Description
Public	Can register, log in, and browse published events, categories, routes, weather information, and published results.
Participant	Can manage a personal profile, enrol in event categories, cancel own enrolments, and view personal results.
Organiser	Can create and manage events owned by the organiser, including categories, enrolments, routes, weather information, and results.
Admin	Can manage and view all users and system resources.

B.3 Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new participant or organiser account.	Public	{ firstName, lastName, email, password, role, phoneNumber }	201 Created with the new user profile and authentication token.

POST	/api/auth/login	Authenticates a registered user.	Public	{ email, password }	200 OK with { token, expiresAt, user } .
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B.4 User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Retrieves the profile of the currently signed-in user.	Participant, Organiser, Admin	None	200 OK with the current user's profile.
PUT	/api/users/me	Updates the currently signed-in user's profile.	Participant, Organiser, Admin	{ firstName, lastName, phoneNumber }	200 OK with the updated profile.
GET	/api/users/{userId}	Retrieves a user profile.	Admin; limited access for an organiser	None	200 OK with a safe user profile, or 404 Not Found .
GET	/api/users/{userId}/results	Retrieves a participant's performance history.	Participant, Admin, or relevant Organiser	None	200 OK with the participant's results and event details.

B.5 Event Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Browses published events. Optional filters may	Public	None; optional query parameters	200 OK with a collection of events.

		include city, province, event type, date, and status.			
GET	/api/events/{eventId}	Retrieves the details of one event.	Public	None	200 OK with event details, or 404 Not Found .
POST	/api/events	Creates a new event.	Organiser, Admin	{ name, description, eventType, venue, city, province, eventData, registrationOpen, registrationClose, status }	201 Created with the new event.
PUT	/api/events/{eventId}	Updates an existing event.	Owning Organiser, Admin	Event details as used when creating an event	200 OK with the updated event.
DELETE	/api/events/{eventId}	Deletes or cancels an event according to the system rules.	Owning Organiser, Admin	None	204 No Content , or 409 Conflict if deletion is not permitted.
GET	/api/organisers/{userId}/events	Lists the events belonging to an organiser.	Owning Organiser, Admin	None	200 OK with an event collection.

B.6 Category Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories for	Public	None	200 OK with a category

	s	an event.			collection.
GET	/api/categories/{categoryId}	Retrieves the details of one category.	Public	None	200 OK with category details, or 404 Not Found .
POST	/api/events/{eventId}/categories	Adds a category to an event.	Owning Organiser, Admin	{ name, distanceKm, entryFee, maximumEntries }	201 Created with the new category.
PUT	/api/categories/{categoryId}	Updates a category.	Owning Organiser, Admin	{ name, distanceKm, entryFee, maximumEntries }	200 OK with the updated category.
DELETE	/api/categories/{categoryId}	Deletes a category when it has no enrolments.	Owning Organiser, Admin	None	204 No Content , or 409 Conflict if enrolments exist.

B.7 Event Enrolment Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments/me	Lists the current participant's enrolments.	Participant, Admin	Optional eventId and status filters	200 OK with the participant's enrolments.
GET	/api/enrolments/{enrolmentId}	Retrieves one enrolment.	Enrolment owner, Event Organiser, Admin	None	200 OK with enrolment details, or 404 Not Found .
POST	/api/categories/{categoryId}/enrolments	Enrols the signed-in participant in a category.	Participant	{ } or optional participant confirmation details	201 Created with the enrolment and race number.

PUT	/api/enrolments/{enrolmentId}/cancel	Cancels an enrolment within the permitted cancellation period.	Enrolment owner, Admin	None	200 OK with the cancelled enrolment, or 409 Conflict .
GET	/api/events/{eventId}/enrolments	Lists enrolments for an organiser's event.	Owning Organiser, Admin	Optional status filter	200 OK with an enrolment collection.
PUT	/api/enrolments/{enrolmentId}/status	Changes the status of an enrolment.	Owning Organiser, Admin	{ status }	200 OK with the updated enrolment.

B.8 Results Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/results	Displays published results for an event.	Public	Optional categoryId and participantId filters	200 OK with an ordered results collection.
GET	/api/results/{resultId}	Retrieves one result.	Public	None	200 OK with result details, or 404 Not Found .
GET	/api/enrolments/{enrolmentId}/result	Retrieves the result connected to an enrolment.	Enrolment owner, Event Organiser, Admin	None	200 OK with the result, or 404 Not Found .
POST	/api/enrolments/{enrolmentId}/result	Records a result for a completed enrolment.	Owning Organiser, Admin	{ finishPosition, finishTimeSeconds, resultStatus }	201 Created with the new result.

PUT	/api/results/{resultId}	Corrects or updates a result.	Owning Organiser, Admin	{ finishPosition, finishTimeSeconds, resultStatus }	200 OK with the updated result.
DELETE	/api/results/{resultId}	Deletes an incorrectly recorded result.	Owning Organiser, Admin	None	204 No Content .

B.9 Route and Weather Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/route	Retrieves route and map information for an event.	Public	None	200 OK with route details, or 404 Not Found .
PUT	/api/events/{eventId}/route	Creates or replaces route information for an event.	Owning Organiser, Admin	{ startLocation, finishLocation, distanceKm, routeDescription , mapUrl }	200 OK with the saved route.
GET	/api/events/{eventId}/weather	Retrieves the latest weather forecast for an event.	Public	Optional forecastDate query parameter	200 OK with forecast information.
POST	/api/events/{eventId}/weather	Stores or refreshes a weather forecast.	Owning Organiser, Admin	{ forecastDate, temperatureC, conditions, windSpeedKmh, rainChancePercent }	201 Created with the saved forecast.

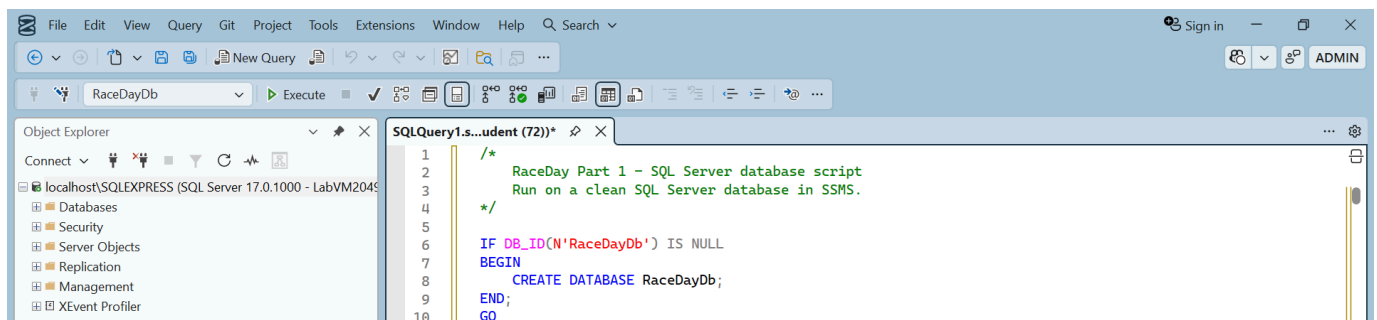
DELETE	/api/weather/{weatherForecastId}	Deletes an obsolete weather forecast.	Owning Organiser, Admin	None	204 No Content .
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B.10 General API Response Codes

Status Code	Meaning
200 OK	The request was successful.
201 Created	A new resource was successfully created.
204 No Content	The request was successful and no response body is required.
400 Bad Request	The request contains invalid or incomplete data.
401 Unauthorized	The user has not authenticated.
403 Forbidden	The user is authenticated but does not have permission.
404 Not Found	The requested resource does not exist.
409 Conflict	The request conflicts with a business rule or existing record.
500 Internal Server Error	An unexpected server error occurred.

SECTION C

proof



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```
USE RaceDayDb;  
GO  
  
/* Drop objects in dependency order so the script can be rerun during testing. */  
DROP TABLE IF EXISTS dbo.WeatherForecasts;
```

100 %
No issues found

Ln: 188, Ch: 1 | SPC | CRLF | Windows 1252

Messages

(4 rows affected)

(3 rows affected)

(6 rows affected)

(3 rows affected)

(3 rows affected)

(4 rows affected)

(2 rows affected)

100 %
No issues found

Ln: 17, Ch: 1 | TABS | MIXED | UTF-8 with BOM

Query executed successfully.

localhost\SQLEXPRESS (17.0 ... | LabVM2049939\Student (72) | RaceDayDb | 00:00:00 | Row: 0, Col: 0 | 0 rows

Ready

11°C Cloudy

Search

10:11 PM 9/4/2026